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CQF Exam One

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Instructions

All questions must be attempted. Requested mathematical workings and coding must be provided to obtain maximum credit. CQF material may be referred to. You are not allowed to discuss the exam work with anyone else. Questions must be answered **in your own words**.

This examination tests your mathematical understanding and ability to evaluate solutions. Some questions provide 'AI-computed solutions' that **may/may not** contain errors—your task is to verify (recompute), then correct and extend them by answering the exact enumerated questions.

Upload **two files**:
1. `E1_YOURNAME_REPORT.pdf` (or `.html`) (or `.docx`)
2. `E1_YOURNAME_CODE.zip` containing all supporting files

Task 1 is computed on the investment universe of the following risky assets. The dependence structure (correlation matrix) is assumed constant.

Asset	$\boldsymbol{\mu}$	$\boldsymbol{\sigma}$
A	0.02	0.05
B	0.07	0.12
C	0.15	0.17
D	0.20	0.25

$$\boldsymbol{R} = \begin{pmatrix} 1 & 0.3 & 0.3 & 0.3 \\ 0.3 & 1 & 0.6 & 0.6 \\ 0.3 & 0.6 & 1 & 0.6 \\ 0.3 & 0.6 & 0.6 & 1 \end{pmatrix}$$

Task 1. Consider **the min variance portfolio with a target return** m.

$$\underset{\boldsymbol{w}}{\operatorname{argmin}} \ \frac{1}{2}\boldsymbol{w}'\boldsymbol{\Sigma}\boldsymbol{w} \qquad \text{subject to} \quad \boldsymbol{w}'\mathbf{1} = 1, \qquad \boldsymbol{w}'\boldsymbol{\mu} = m$$

(a) Formulate the Lagrangian and state the first-order conditions. No further derivation work required.

(b) Compute the intermediate values A, B, C, D as reviewed in CQF Lecture on Portfolio Optimisation.

(c) An AI system computed the following solution for $m = 4.5\%$.

AI-Computed Solution

Computed optimal weights:

$w_A = 78.51\%, \quad w_B = 5.39\%, \quad w_C = 13.36\%, \quad w_D = 2.75\%$

Portfolio risk: $\sigma_\Pi = 5.84\%$

Verify this solution by confirming the constraints are satisfied, and computing the portfolio variance independently. Provide the computation working.

(d) Provide at least two explanations to why Asset A receives the largest allocation $w_A = 78.51\%$ despite having the lowest expected return.
Hint: you may consider volatility level, correlation, the level of portfolio target return.

Task 2. Consider a portfolio of three assets with the following risk characteristics:

Asset	$\boldsymbol{\mu}$	$\boldsymbol{\sigma}$	$\boldsymbol{w}$
1	0%	30%	50%
2	0%	20%	20%
3	0%	15%	30%

$$\boldsymbol{\rho} = \begin{pmatrix} 1 & 0.8 & 0.5 \\ 0.8 & 1 & 0.3 \\ 0.5 & 0.3 & 1 \end{pmatrix}$$

VaR and ES sensitivities are computed *with regard to* each individual asset i, and notation $()_i$ refers to i-th element. Letters in bold signify vector variables. Factor $z_\alpha = \Phi^{-1}(1-c) = \Phi^{-1}(0.01) \approx -2.326$.

$$\frac{\partial \mathrm{VaR}(\boldsymbol{w})}{\partial w_i} = \mu_i + z_\alpha \cdot \frac{(\boldsymbol{\Sigma w})_i}{\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma w}}}$$

$$\frac{\partial \mathrm{ES}(\boldsymbol{w})}{\partial w_i} = \mu_i - \frac{\phi(z_\alpha)}{1-c} \cdot \frac{(\boldsymbol{\Sigma w})_i}{\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma w}}}$$

confidence $c = 99\%$, and $\phi(\cdot)$ is the standard normal PDF. Signs in these formulae are as intended.

AI-Computed Solution

Asset	Weight	$\frac{\partial \mathrm{VaR}}{\partial w_i}$	$\frac{\partial \mathrm{ES}}{\partial w_i}$
1	50%	−0.6839	−0.7835
2	20%	−0.3868	−0.4431
3	30%	−0.2207	−0.2529

(a) Verify the computation of at least two sensitivities of your choice.

(b) Why are all sensitivities negative? What does a negative $\frac{\partial \mathrm{VaR}}{\partial w_i}$ mean for portfolio management?

(c) Asset 1 has the highest volatility (30%) but also the largest allocation (50%). Why does it have the most negative sensitivity?

(d) Comparing the VaR vs ES, why is $|\frac{\partial \mathrm{ES}}{\partial w_i}| > |\frac{\partial \mathrm{VaR}}{\partial w_i}|$ for all assets? You can either provide the strictly mathematical reason or fall back on the definitions of VaR and ES.

Task 3. VaR Backtesting

As a market risk analyst, you are asked to compute the VaR backtesting for **S&P500** from 2025-01-01 to 2026-01-15 according to the following methodology. Computation coding and plot quality will be evaluated.

(a) Report the count and percentage of VaR breaches.

(b) Present a plot which identifies the breaches visually (marks can be crosses).

(c) Report traffic-light ranges for breaches (green / yellow / red).

Data file provided, you need to apply start and end dates. Closing Prices are real life but we don't recommend downloading the data separately.

- Compute the rolling standard deviation σ_t from 21 daily returns.
- Timescale of that σ_t remains 'daily' regardless of how many returns are in the sample. To make projection, use the additivity of variance $\sigma^{10D} = \sqrt{\sigma_t^2 \cdot 10}$.
- Use a simplified formula:

$$\text{10D VaR}_t = \text{Factor} \cdot \sigma_t \cdot \sqrt{10}.$$

- A breach occurs when the forward 10D return is below the VaR_t quantity.

$$r^{10D} < \mathrm{VaR}_t \qquad \text{breach, both numbers are negative.}$$

  This forward-looking return is computed as $\ln(S_{t+10}/S_t)$.
  If you explore $\ln(S_{t+11}/S_{t+1})$ in addition, make the assumption clear and results separated.

The number of exceptions satisfies $B \sim \mathrm{Binomial}(T, 0.01)$. Assume $B = b$ exceptions realised in our data.

- Risk traffic-light zones are computed by *binomial percentiles* q_p and depend on the number of eligible comparisons T, on which VaR_t is compared to the 10D return.
- Green zone is the most restrictive range of breaches $0 < b \leq q_{p_G}$, probability $p_G = 0.95$. Yellow zone is $q_{p_G} < b \leq q_{p_Y}$, red zone if breaches are too many $b > q_{p_Y}$. Take $p_Y = 0.9999$.

  Example: $T = 250$ comparisons give Yellow zone of $5 - 9$ breaches but your numbers will differ.

Task 4. Reimplement the backtesting but now utilise the **EWMA volatility forecast** to compute 10D VaR_t. Set $\lambda = 0.72$. The dataset variance initializes the computation.

$$\sigma^2_{t+1\,|\,t} = \lambda\,\sigma^2_{t\,|\,t-1} + (1-\lambda)\,r_t^2$$

(a) Report the count and percentage of VaR breaches (for this EWMA method).

(b) Present a plot which identifies the breaches visually (marks can be crosses).

(c) Among the three approaches studied (Rolling SD, EWMA, HistSim) which exhibits cases of *best* and *worst* adaptivity to volatility? Provide a brief discussion explaining the responsiveness for each case.

Notes

EWMA computation does not rely on a rolling window; please refer to CQF material and remember it's your task to combine the knowledge here.

The answer to (c) must be supported by mathematical reasoning. Each case can take two paragraphs.

Task	**Marks**
Task 1. MVO-generated allocations	25
Task 2. VaR and ES sensitivities	20
Task 3. VaR backtesting	25
Task 4. EWMA method	30

END OF TASKS – CQF EXAM ONE

Report Preparation

You must prepare **a report that integrates workings, numerical answers and plots** in the order of tasks. The report may be submitted as a PDF, HTML, or Word file.

You do not need to typeset a full report in LaTeX, but you may do so if you prefer.

Most delegates work within a single Python notebook, but this is not required. You are not limited to Python language too. You may prepare your report in parts and then combine them into one PDF file. For example, you might use separate notebooks/code files for different tasks, complete derivations in LaTeX or by hand, or summarise numerical answers in Word.

Report Guidelines

1. **If drafting from a Python notebook:** Remove unnecessary output and unused code. Add mathematical explanations using Markdown/LaTeX. Save `.ipynb` file to HTML first.
2. **If drafting from Word or LaTeX:** Include the requested maths, numerical solutions and plots. You may insert code at your choice—either after each question or collected in an appendix.
3. **Handwritten workings:** Scanned handwritten derivations and numerical work (where feasible) are acceptable without penalty, provided they are legible.
4. **Where a task specifies a particular format:** results in a table, a specific plot, follow those instructions.

What Is NOT a Report

- Raw "code + output" Python printouts do not constitute a report. They lack mathematical explanation and do not address the specific questions asked.
- Submission of only Python code with comments, or an unadapted Python notebook, will receive a **significant mark deduction**. You may also be asked to resubmit work that meets the exam requirements.
- Excel-only work is possible for Exam One but **not recommended**. Spreadsheet printouts with computational results and copied formulae will not be considered a proper report.

Making operational sense of the exam questions and implementing them in code is part of your individual exam work. Please make good use of lectures, solutions, and labs/tutorials. The tutor is unable to confirm formulae, discuss your numerical answers, or provide hints beyond those already given.

Required Submissions (Two Files)

1. `E1_YOURNAME_REPORT.pdf` (or `.html`) (or `.docx`)
2. `E1_YOURNAME_CODE.zip` containing all supporting files

The ZIP file should include:

- Original and runnable Python notebook, Python or other code files.
- Signed declaration (if not included in the PDF report) and any other supporting files.
- Do **not** use RAR format—use ZIP only.
- Do **not** place the report inside the ZIP file. The report PDF/HTML should be uploaded as a separate file.

Submissions that do not follow these instructions will require additional processing time.